The threshold identity

$\|\mathfrak{T}_0\| = 1/\sqrt{\lambda_{\min}(C)}$ is the energy split on the quotient; exact saturation on PD is not

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Abstract

Round 409 printed $\|\mathfrak{T}_0\| = 1.00000$ on the positive-definite windows $k_z = 42, 130$. That looked like a combinatorial identity. It is not. The exact lemma is the dictionary

$$\mathfrak{T}_0^*\mathfrak{T}_0 \;=\; C^{-1}, \qquad \|\mathfrak{T}_0\| \;=\; 1/\sqrt{\lambda_{\min}(C)},$$

and the energy split on the quotient $\mathcal{P}_{<d_0}/\mathcal{K}_0$: $\|\mathfrak{T}_0\| \le 1$ if and only if every min-norm interpolant carries at least half its dual-weighted energy on Y. On PD windows, $C_{\min} = 1+\varepsilon$ with $\varepsilon \in [10^{-8}, 10^{-4}]$, so $\|\mathfrak{T}_0\|$ sits $O(\varepsilon)$ below 1. No saturating polynomial with exact half-half split exists on the tested family. The excess direction of a P1 window is C's lowest eigenvector (which *is* $\mathfrak{T}_0$'s top singular vector), not the Fourier hole-Nyquist face of r410. No RH claim.

Status. Lemmas 1–2 are proved (finite identities over $\mathbb{Q}$ and a machine-exact dictionary). Exact $\|\mathfrak{T}_0\| = 1$ on PD, a combinatorial saturator, and "$k = 37$ is a named Christoffel-weight hole" are **refuted**. Ausgang `ENERGY_SPLIT_EXACT / SATURATION_REFUTED`. No L^* claim (this is a finite dual-coordinate dictionary, not the L^* subordination). No $R^\dagger$ claim. No RH claim.

1 The quotient

Write $d_0 = N_w - 3$ and $\mathcal{K}_0 = \ker(\mathrm{ev}_Y)$ inside $\mathcal{P}_{<d_0}$, so $\mathcal{K}_0 = P_Y \cdot \mathcal{P}_{<d_0-|Y|}$. $\mathfrak{T}_0$ is the min-norm interpolant $Y \to X$ in the dual weights $u^\vee$ (r409). By construction, $\mathfrak{T}_0(W_Y^{1/2} q|_Y) = W_X^{1/2} q^*|_X$ where q^* is the X-orthogonal projection of q onto $\mathcal{K}_0^\perp$. On $\mathcal{K}_0$ the Y-energy vanishes and the split does not apply.

Lemma 1 (Dictionary and projection, over $\mathbb{Q}$)**.** *On the five-node toy of r409, $\mathfrak{T}_0^*\mathfrak{T}_0 = C^{-1}$ to $4.6 \cdot 10^{-17}$ and $\|\mathfrak{T}_0\| = 1/\sqrt{\lambda_{\min}(C)}$. P_Y has Y-energy 0. After subtracting the $\mathcal{K}_0$-component in the X-inner product, $\mathfrak{T}_0(s_Y q|_Y) = s_X q_{\min}|_X$ to $1.3 \cdot 10^{-15}$. The mutants $u^\vee = 1/u$ and $d_0 - 1$ break the identity.*

Lemma 2 (Energy split)**.** *For $q \in \mathcal{K}_0^\perp$ write $v = W_Y^{1/2} q|_Y$. Then*

$$\frac{\|q\|^2_{X,u^\vee}}{\|q\|^2_{Y,u^\vee}} \;=\; \frac{v^T C^{-1} v}{v^T v}.$$

Hence $\|\mathfrak{T}_0\| \le 1 \iff C \succeq I \iff$ every such q satisfies $\|q\|_X \le \|q\|_Y$. A raw Chebyshev T_k is not in the quotient: on FRAME-A, T_1 has energy ratio $2.73 > \|\mathfrak{T}_0\|^2 = 1.17$.

Lemma 3 (Machine dictionary)**.** *On FRAME-A, $\|\mathfrak{T}_0^*\mathfrak{T}_0 - C^{-1}\|_F / \|C^{-1}\|_F = 6.64 \cdot 10^{-14}$ and $\|\mathfrak{T}_0\| = 1.080138437324 = 1/\sqrt{C_{\min}}$ to 10^{-14}, $C_{\min} = 0.857119$. The top right singular vector of $\mathfrak{T}_0$ equals C's lowest eigenvector (correlation 1). The Fourier hole-Nyquist face v_F of r410 has correlation 0.386 and $\|\mathfrak{T}_0 v_F\| = 0.656538 < 1.080$. The μ-OP profile $B^T v$ peaks at index 165/181 (deep, not a low-degree T_k).*

2 Saturation is not exact

Proposition 1 (PD gap, not an identity)**.** *On* $k_z = 42$*,* $C_{\min} - 1 = +4.565 \cdot 10^{-8}$ *and* $\|\mathfrak{T}_0\| - 1 = -2.283 \cdot 10^{-8}$*. On* $k_z = 130$*,* $C_{\min} - 1 = +4.90 \cdot 10^{-9}$*. On core-*42 *the* 14 *vacuous windows have* $C_{\min} - 1 \in [1.33 \cdot 10^{-7}, 7.54 \cdot 10^{-5}]$*, none equal to* 0*. There is no tested combinatorial* q *with exact* $\|q\|_X = \|q\|_Y$ *on the quotient. The printed* 1.00000 *of r409 is* $1/\sqrt{1+\varepsilon}$*.*

Proposition 2 ($k = 37$ is an eigenvalue crossing)**.** *Along the* θ*-ordered prefix of FRAME-A holes,* $k = 36$ *has* $C_{\min} - 1 = +6.82 \cdot 10^{-5}$ *(*$n_C = 0$*) and* $k = 37$ *has* $C_{\min} - 1 = -9.23 \cdot 10^{-5}$ *(*$n_C = 1$*,* $\|\mathfrak{T}_0\| = 1.000046$*). The last-added Christoffel weight is* $K = 6.94$*, not a collapse. Leave-one-out from the* 37*-prefix restores* $n_C = 0$ *for several holes, so the birth is not a single named atom. Permute (same* Y*) already exceeds at prefix* $k = 10$*; its* $K_{\min} = 0.148$ *appears only at full* Y *(r410).*

3 Kills

Permute: $n_C = 20$, $\|\mathfrak{T}_0\| = 7.42$, $K_{\min} = 0.148$ (Christoffel collapse of the μ-chain; the dictionary still holds). Scramble: $n_C = 21$, $\|\mathfrak{T}_0\| = 16.66$, $C_{\min} = 0.0036$. Periodic 1010 at $S = 20$: $\|\mathfrak{T}_0\| = 165$, $n_C = 3$. Living χ_3-9: $\|\mathfrak{T}_0\| = 0.99956 < 1$; dead χ_3-15: $\|\mathfrak{T}_0\| = 1.00092 > 1$ (the split holds; death is edge, r401).

4 What this reduces, and what it does not

P1 is exactly one excess direction on the quotient: exactly one $q \in \mathcal{K}_0^\perp$ with $\|q\|_X > \|q\|_Y$, equivalently $\#\{\lambda(C) < 1\} = 1$, equivalently exactly one singular value of $\mathfrak{T}_0$ above 1. That is r407 in energy language. It is not an exact half-filling identity, not a combinatorial saturator, and not L^*. The mincut does not move.

Machine check

Standalone over $\mathbb{Q}$: $\mathfrak{T}^*\mathfrak{T} = C^{-1}$, projection identity, kernel P_Y, two mutants. Construction: FRAME-A dictionary and Fourier miss, $k_z = 42$ gap, $k = 36/37$ crossing, permute / scramble / 1010 / χ. Discovery census: `threshold_identity_probe.py` (`--smoke` 17/17 / full 20/20), SPEC_SHA `090cbc12b41cb27d`, wall 23.4 s. Exit line: `THRESHOLD IDENTITY VERIFIED`.

Claim boundary

Finite identities for the dual energy split on the quotient $\mathcal{P}_{<d_0}/\mathcal{K}_0$, and a named refutation of exact saturation. No statement about zeros of $\zeta(s)$, in either direction. No L^* claim. No $R^\dagger$ claim. No RH claim.

References

[1] S. Hamann, *The dual intertwiner*, `rh/problem/dual_intertwiner.tex`, August 2026.

[2] S. Hamann, *The Borodin–Birkhoff intertwiner*, `rh/problem/borodin_birkhoff_intertwiner.tex`, August 2026.

[3] S. Hamann, *The hole-Nyquist defect*, `rh/problem/hole_nyquist.tex`, August 2026.